QUESTION ANALYSIS

# EEE455

VLSI Design and Testing

# Question Analysis

**Sessions 2019–20 to 2020–21**

*2 papers, 53 questions, 90 formulas*

INSIDE

• At a glance • Heat map • Priority list • 12 topic pages • Derivations • Repeats • Styles • Study order

Prepared by Nur Mohammad

October 2026

# Contents


**1 At a glance** **3**

**2 Heat map** **3**

**3 Priority list** **4**

**4 Topic pages** **5**

4.1 MOS transistor theory . . . 5
4.2 CMOS inverter . . . 6
4.3 Fabrication . . . 7
4.4 Delay and logical effort . . . 8
4.5 Power . . . 11
4.6 Interconnect . . . 12
4.7 Scaling . . . 12
4.8 Combinational logic styles . . . 13
4.9 Sequential circuits and timing . . . 14
4.10 Datapath . . . 16
4.11 Memories and arrays . . . 17
4.12 Verilog and design flow . . . 17

**5 Derivations** **18**

**6 Repeat tracker** **18**

**7 Question style** **18**

**8 Study order** **18**




Free shared copy · Nur Mohammad

# 1. At a glance:

This analysis covers **2 papers** (2019-20, 2020-21), with 53 questions and 294 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

### The five topics with the most marks:

| Topic | Marks | Share | Questions | Top subtopics (marks) |
|---|---|---|---|---|
| Combinational logic styles | 49 | 17 % | 10 | Stick diagrams (12); Static gates (9); Dynamic logic (9) |
| Delay and logical effort | 46 | 16 % | 8 | Path delay (25); Gate delay (7); Input ordering (5) |
| Sequential circuits and timing | 36 | 12 % | 6 | Setup and hold (10); Flip-flops (10); Hold time (6) |
| Fabrication | 30 | 10 % | 6 | Lithography (16); Etching (6); Cleaning (6) |
| Power | 28 | 10 % | 5 | Dynamic power (16); Glitches (6); Leakage (6) |

A formula name in bold is a key formula.

# 2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

| Topic and subtopic | 2019-20 | 2020-21 | All |
|---|---|---|---|
| **MOS transistor theory** | **13** | **4** | **17** |
| I-V equations | **8** | · | 8 |
| Leakage and DIBL | **5** | · | 5 |
| C-V characteristics | · | **4** | 4 |
| **CMOS inverter** | **7** | **13** | **20** |
| DC characteristics | **7** | · | 7 |
| Skew | · | **7** | 7 |
| Beta ratio | · | **6** | 6 |
| **Fabrication** | **20** | **10** | **30** |
| Lithography | **8** | **8** | 16 |
| Etching | **6** | · | 6 |
| Cleaning | **6** | · | 6 |
| Diffusion and implantation | · | **2** | 2 |
| **Delay and logical effort** | **25** | **21** | **46** |
| Path delay | **13** | **12** | 25 |
| Gate delay | **7** | · | 7 |
| Input ordering | **5** | · | 5 |
| Elmore delay | · | **5** | 5 |
| Sizing | · | **4** | 4 |
| **Power** | **16** | **12** | **28** |
| Dynamic power | **4** | **12** | 16 |
| Glitches | **6** | · | 6 |
| Leakage | **6** | · | 6 |
| **Interconnect** | **10** | · | **10** |
| Crosstalk | **7** | · | 7 |
| Sheet resistance | **3** | · | 3 |
| **Scaling** | **11** | · | **11** |

| Topic and subtopic | 2019-20 | 2020-21 | All |
|---|---|---|---|
| Moore's law | **6** | · | 6 |
| Scaling rules | **5** | · | 5 |
| **Combinational logic styles** | **25** | **24** | **49** |
| Stick diagrams | **6** | **6** | 12 |
| Static gates | **9** | · | 9 |
| Dynamic logic | **6** | **3** | 9 |
| Pseudo-nMOS | · | **8** | 8 |
| Pass transistors | · | **7** | 7 |
| CVSL | **4** | · | 4 |
| **Sequential circuits and timing** | **10** | **26** | **36** |
| Setup and hold | · | **10** | 10 |
| Flip-flops | · | **10** | 10 |
| Hold time | **6** | · | 6 |
| FSM | · | **6** | 6 |
| Cycle time | **4** | · | 4 |
| **Datapath** | · | **7** | **7** |
| Bus architectures | · | **7** | 7 |
| **Memories and arrays** | **13** | **13** | **26** |
| SRAM | **6** | **10** | 16 |
| ROM | **7** | · | 7 |
| PLA | · | **3** | 3 |
| **Verilog and design flow** | · | **14** | **14** |
| Verilog | · | **8** | 8 |
| Design flow | · | **6** | 6 |

# 3. Priority list:

**Must know:** Combinational logic styles (49), Delay and logical effort (46), Sequential circuits and timing (36), Fabrication (30), Power (28), Memories and arrays (26), CMOS inverter (20), MOS transistor theory (17)

**Likely:** Verilog and design flow (14), Scaling (11), Interconnect (10), Datapath (7)

**Less often:** none

**The rule:** Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.


Free shared copy · Nur Mohammad

# 4. Topic pages:

## 4.1. MOS transistor theory:

17 marks, 3 questions, in 2 of 2 papers. Book: Weste Ch. 2; Kang Ch. 3.

Most often theory (2 of 3). Subtopics: Leakage and DIBL; I-V equations; C-V characteristics.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Leakage and DIBL | **DIBL** | $V_t = V_{t0} - \eta V_{DS}$ | 19-20 1(d) |
| | **Subthreshold current** | $I_{DS} = I_{ds0}\, e^{\frac{V_{GS}-V_t+\eta V_{DS}}{nv_T}} \left(1 - e^{-V_{DS}/v_T}\right)$ | 19-20 1(d) |
| | Subthreshold slope | $S = nv_T \ln 10$ | 19-20 1(d) |
| I-V equations | Oxide capacitance | $C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$ | 19-20 2(a), 20-21 1(a) |
| | Gain factor | $\beta = \mu C_{ox} \frac{W}{L}$ | 19-20 2(a) |
| | **Current, linear** | $I_{DS} = \beta \left(V_{GS} - V_t - \frac{V_{DS}}{2}\right) V_{DS}$ | 19-20 2(a) |
| | **Current, saturation** | $I_{DS} = \frac{\beta}{2}(V_{GS} - V_t)^2$ | 19-20 2(a) |
| C-V characteristics | **Gate capacitance** | $C_0 = C_{ox} W L$ | 20-21 1(a) |
| | Gate capacitance, cutoff | $C_{gb} \approx C_0,\ C_{gs} = C_{gd} = 0$ | 20-21 1(a) |
| | Gate capacitance, linear | $C_{gs} = C_{gd} = \frac{1}{2}C_0,\ C_{gb} = 0$ | 20-21 1(a) |
| | Gate capacitance, saturation | $C_{gs} = \frac{2}{3}C_0,\ C_{gd} = C_{gb} = 0$ | 20-21 1(a) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Leakage and DIBL** | | | | |
| 1 | 19-20 1(d) | 5 | theory | Explain the following non-idealities of MOSFET- (i) Drain induced barrier lowering, (ii) subthreshold leakage. What are the effects of temperature on these non-idealities? |
| **I-V equations** | | | | |
| 2 | 19-20 2(a) | 8 | derivation | Find the equations and I-V characteristics curve of MOSFET current using long-channel model. |
| **C-V characteristics** | | | | |
| 3 | 20-21 1(a) | 4 | theory | Explain the C-V characteristics of intrinsic MOS gate capacitance in different regions. |


Free shared copy · Nur Mohammad

## 4.2. CMOS inverter:

20 marks, 3 questions, in 2 of 2 papers. Book: Weste Ch. 2; Kang Ch. 5, 6.

Most often theory (3 of 3). Subtopics: DC characteristics; Beta ratio; Skew.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| DC characteristics | Operating point | $I_{Dn} = \lvert I_{Dp} \rvert$ | 19-20 3(a) |
| | **Switching threshold** | $V_M = \frac{V_{tn} + r(V_{DD} - \lvert V_{tp} \rvert)}{1+r},\ r = \sqrt{\beta_p/\beta_n}$ | 19-20 3(a) |
| | Unity-gain inputs | $V_{IL} = \frac{3V_{DD} + 2V_t}{8},\ V_{IH} = \frac{5V_{DD} - 2V_t}{8}$ | 19-20 3(a) |
| | Noise margins | $NM_L = V_{IL} - V_{OL},\ NM_H = V_{OH} - V_{IH}$ | 19-20 3(a), 20-21 2(b) |
| Beta ratio | **Switching threshold** | $V_M = \frac{V_{DD} - \lvert V_{Tp} \rvert + V_{Tn}\sqrt{\beta_n/\beta_p}}{1+\sqrt{\beta_n/\beta_p}}$ | 20-21 2(b) |
| | Symmetric CMOS $V_{IL}$ | $V_{IL} = \frac{3V_{DD} + 2V_T}{8},\ V_{IH} = \frac{5V_{DD} - 2V_T}{8}$ | 20-21 2(b) |
| Skew | HI-skew inverter | $g_u = \frac{5}{6},\ g_d = \frac{5}{3}$ | 20-21 3(d) |
| | LO-skew inverter | $g_u = \frac{4}{3},\ g_d = \frac{2}{3}$ | 20-21 3(d) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **DC characteristics** | | | | |
| 1 | 19-20 3(a) | 7 | theory | Using I-V characteristics curve of MOSFET, find DC characteristics of CMOS inverter. |
| **Beta ratio** | | | | |
| 2 | 20-21 2(b) | 6 | theory | Briefly explain what happens to the transfer characteristics and noise margins when beta ratio is increased in a CMOS inverter pair. |
| **Skew** | | | | |
| 3 | 20-21 3(d) | 7 | theory | Explain the inverter width sizing for favoring the rising edge, falling edge, or achieving an unskewed inverter. |


Free shared copy · Nur Mohammad

## 4.3. Fabrication:

30 marks, 6 questions, in 2 of 2 papers. Book: Weste Ch. 3; Plummer Ch. 4, 5, 7, 8, 10.

Most often theory (6 of 6). Subtopics: Etching; Cleaning; Lithography; Diffusion and implantation.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Etching | Plasma dissociation | $CF_4 + e^- \rightarrow CF_3 + F + e^-$ | 19-20 1(a) |
| | **Silicon etch** | $Si + 4F \rightarrow SiF_4 \uparrow$ | 19-20 1(a) |
| | Oxide etch (dry) | $SiO_2 + 4F \rightarrow SiF_4 \uparrow + O_2$ | 19-20 1(a) |
| | Oxide etch (wet) | $SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O$ | 19-20 1(a) |
| | **Adding $O_2$** | $O + CF_3 \rightarrow COF_2 + F$ | 19-20 1(a) |
| | **Adding $H_2$** | $H + F \rightarrow HF$ | 19-20 1(a) |
| | Selectivity | $S = \frac{r_{film}}{r_{under}}$ | 19-20 1(a) |
| Cleaning | **RCA clean** | SC-1: $NH_4OH$:$H_2O_2$:$H_2O$ = 1:1:5; HF 1:50; SC-2: HCl:$H_2O_2$:$H_2O$ = 1:1:6 | 19-20 1(b) |
| Lithography | Lithography resolution | $W_{min} = k_1 \frac{\lambda}{NA}$ | 19-20 1(c) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Etching** | | | | |
| 1 | 19-20 1(a) | 6 | theory | Briefly discuss the chemistries involved in in *[sic: should be "involved in"]* etching. Also explain the effect of adding $H_2$ and $O_2$ on etching rate. |
| **Cleaning** | | | | |
| 2 | 19-20 1(b) | 6 | theory | What is RCA clean? What are the steps? What contaminants can be removed in this process? |
| **Lithography** | | | | |
| 3 | 19-20 1(c) | 8 | theory | Define photolithography. briefly describe the major steps involved in photolithography. |
| 4 | 20-21 5(c) | 2 | theory | Explain the following: (i) positive mask, (ii) negative mask, (iii) positive photoresist and (iv) negative photoresist. |
| 5 | 20-21 5(d) | 6 | theory | Describe the process sequence for patterning $SiO_2$ as illustrated in the figure 5(d). |
| **Diffusion and implantation** | | | | |
| 6 | 20-21 1(e) | 2 | theory | Differentiate between the diffusion process and the ion implantation process in semiconductor fabrication. |

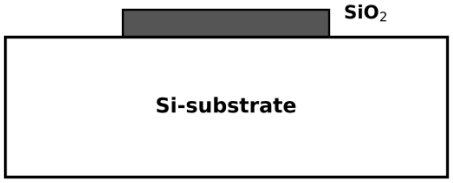



**Figure 1:** 20-21 5(d): Figure 5(d) of the paper (redrawn)


Free shared copy · Nur Mohammad

## 4.4. Delay and logical effort:

46 marks, 8 questions, in 2 of 2 papers. Book: Weste Ch. 4.

Most often numerical (5 of 8). Subtopics: Gate delay; Path delay; Input ordering; Elmore delay; Sizing.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Gate delay | **Elmore delay** | $t_{pd} = \sum_i R_i C_i$ | 19-20 2(b) |
| | NAND3 rising, worst | $t_{pdr} = (9+5h)RC$ | 19-20 2(b) |
| | NAND3 falling, worst | $t_{pdf} = (12+5h)RC$ | 19-20 2(b) |
| | NAND3 contamination | $t_{cdr} = \left(3+\frac{5h}{3}\right)RC,\ t_{cdf} = (9+5h)RC$ | 19-20 2(b) |
| Path delay | **Gate delay** | $d = gh + p$ | 19-20 3(b), 20-21 1(d) |
| | Logical efforts | inv $g=1$; NAND-$n$ $\frac{n+2}{3}$; NOR-$n$ $\frac{2n+1}{3}$; AOI22 2 | 19-20 3(b) |
| | Branching effort | $b = \dfrac{C_{on}+C_{off}}{C_{on}}$ | 19-20 3(b) |
| | **Path effort** | $F = GBH,\ G = \prod g_i,\ B = \prod b_i,\ H = \dfrac{C_{out}}{C_{in}}$ | 19-20 3(b), 20-21 1(d) |
| | **Minimum delay** | $\hat{f} = F^{1/N},\ \hat{D} = NF^{1/N} + P$ | 19-20 3(b) |
| | Gate size | $C_{in,i} = \dfrac{g_i C_{out,i}}{\hat{f}}$ | 19-20 3(b), 20-21 1(d) |
| | **Best stages** | $\hat{N} = \log_4 F$ | 19-20 3(c) |
| | Logical effort, NAND and NOR | $g_{NAND\text{-}n} = \dfrac{n+2}{3},\ g_{NOR\text{-}n} = \dfrac{2n+1}{3}$ | 20-21 1(d) |
| | **Minimum path delay** | $\hat{D} = NF^{1/N} + P,\ \hat{f} = F^{1/N}$ | 20-21 1(d) |
| | Best number of stages | $\hat{N} \approx \log_4 F$ | 20-21 2(c) |
| | FO4 delay | $d_{FO4} = 1\cdot 4 + 1 = 5\tau$ | 20-21 1(d) |
| Input ordering | Input order (NAND2) | outer $p=2$, inner $p=7/3$ | 19-20 3(d) |
| Elmore delay | **Elmore delay** | $t_{D,i} = \sum_k R_{ik} C_k$ | 20-21 2(a) |
| Sizing | Unit inverter | $W_p = \dfrac{\mu_n}{\mu_p} W_n$ | 20-21 3(b) |
| | **Series stack** | $W = k\, W_{unit}$ | 20-21 3(b) |
| | Path resistance | $\sum_{\text{path}} \dfrac{R_{unit} W_{unit}}{W_j} \le R_{unit}$ | 20-21 3(b) |


Free shared copy · Nur Mohammad

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Gate delay** | | | | |
| 1 | 19-20 2(b) | 7 | numerical | Find propagation and contamination delay (for both rising and falling) of 3-input NAND gate while it is driving 4 identical NAND gates. Consider that the gate has equal pull-up and pull-down resistance. |
| **Path delay** | | | | |
| 2 | 19-20 3(b) | 6 | numerical | Find minimum delay for path $A \rightarrow B$ in the logic circuit of Figure 3.1. Also find the gate size to achieve this delay. |
| 3 | 19-20 3(c) | 7 | design | Figure 3.2 shows a portion of a decoder circuit. The inverter at input drives 8 identical 4-input NAND gate with input capacitance y each. The output of the decoder (NAND gate) is connected to a 32-bit word of register and each register bit presents a load of 3-unit sized transistor. Find best number stages (how many inverters should be added between NAND and load?), stage effort, path delay, NAND gate size. |
| 4 | 20-21 1(d) | 5 | numerical | Estimate the minimum delay of the path from $A$ to $B$ in Figure 1(d) and choose transistor sizes to achieve this delay. The initial NAND2 gate may present a load of 8 $\lambda$ of transistor width on the input and the output load is equivalent to 45 $\lambda$ of transistor width. |
| 5 | 20-21 2(c) | 7 | design | Design an 8-input AND gate using FO4 in different configurations. Determine which configuration provides the fastest performance. |
| **Input ordering** | | | | |
| 6 | 19-20 3(d) | 5 | theory | Explain input ordering delay effect. |
| **Elmore delay** | | | | |
| 7 | 20-21 2(a) | 5 | numerical | For the RC circuit shown in Figure 2(a), compute the Elmore delays from input node to node 7 and from input node to node 5 of this circuit in terms of R and C component values. |
| **Sizing** | | | | |
| 8 | 20-21 3(b) | 4 | numerical | For the logic function $Y = \overline{(A + BC) \cdot D}$, choose transistor widths to achieve equal rise and fall resistance as a unit inverter. Assume, $\mu_n = 3\mu_p$ |

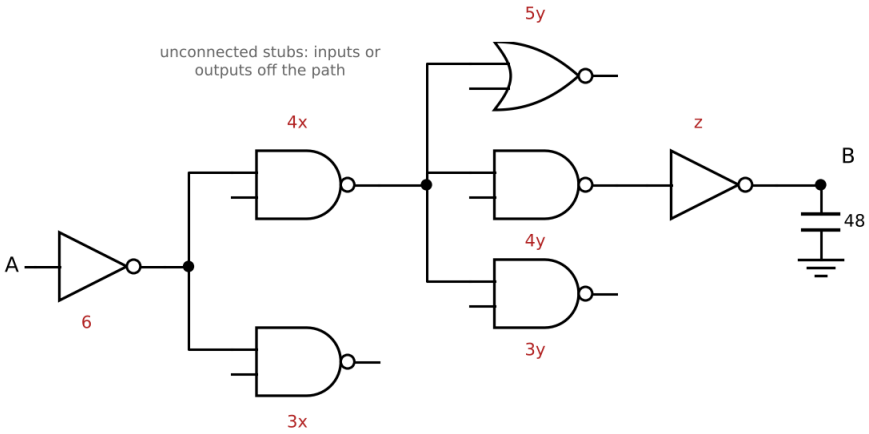



**Figure 2:** 19-20 3(b): Figure 3.1 of the paper (redrawn)


Free shared copy · Nur Mohammad

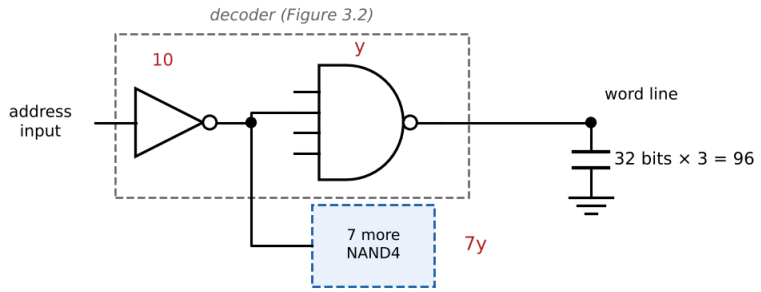



**Figure 3:** 19-20 3(c): Figure 3.2 of the paper (redrawn)

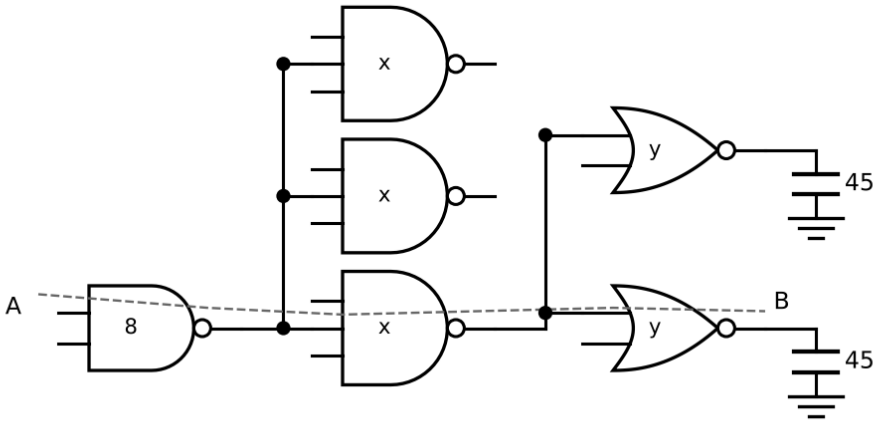



**Figure 4:** 20-21 1(d): Figure 1(d) of the paper (redrawn)

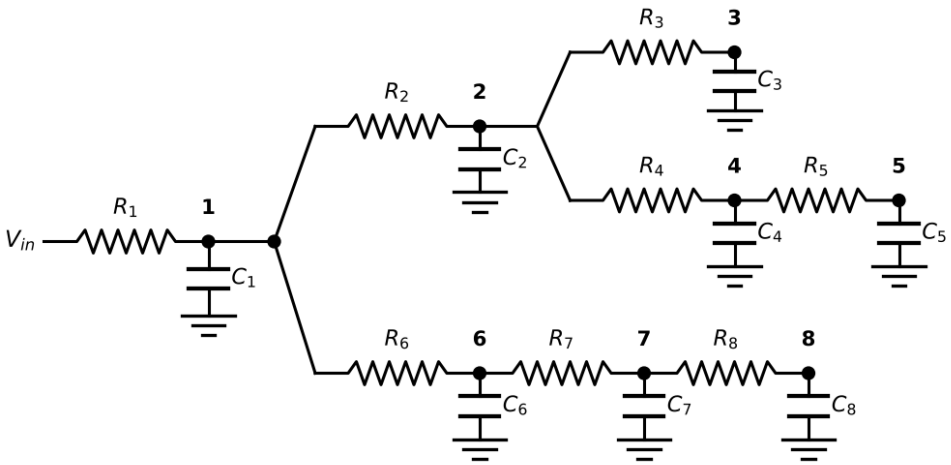



**Figure 5:** 20-21 2(a): Figure 2(a) of the paper (redrawn)


Free shared copy · Nur Mohammad

## 4.5. Power:

28 marks, 5 questions, in 2 of 2 papers. Book: Weste Ch. 5.

Most often theory (3 of 5). Subtopics: Glitches; Dynamic power; Leakage.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Dynamic power | **Dynamic power** | $P_{dyn} = \alpha C V_{DD}^2 f$ | 19-20 4(c) |
| | Capacitance from area | $C = (C/A) \times A$ | 19-20 4(c) |
| | Delay from current | $t_{pd} = \frac{C_L V_{DD}/2}{I_{sat}} = \frac{C_L V_{DD}}{\beta(V_{DD} - V_T)^2}$ | 20-21 3(c) |
| | **Delay vs supply** | $t_{pd} \approx \frac{C_L}{\beta V_{DD}} \propto \frac{1}{V_{DD}}$ | 20-21 3(c) |
| | Energy per output cycle | $E = C_L V_{DD}^2$ | 20-21 3(c) |
| | Energy stored | $E_C = \frac{1}{2} C_L V_{DD}^2$ | 20-21 3(c) |
| | **Dynamic power** | $P_{dynamic} = \alpha C_L V_{DD}^2 f$ | 20-21 2(d) |
| | Static power | $P_{static} = I_{static} V_{DD}$ | 20-21 2(d) |
| Leakage | Stack node voltage | $V_x = \frac{\eta V_{DD}}{1 + 2\eta}$ | 19-20 5(d) |
| | **Stack effect** | $\frac{I_{single}}{I_{stack}} = 10^{\frac{\eta V_{DD}(1+\eta)}{(1+2\eta)S}}$ | 19-20 5(d) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Glitches** | | | | |
| 1 | 19-20 4(b) | 6 | theory | Define glitch? How does it increase power consumption? How it can be reduced? |
| **Dynamic power** | | | | |
| 2 | 19-20 4(c) | 4 | numerical | You are synthesizing a chip composed of random logic with an average activity factor of 0.1. You are using a standard cell process with an average switching capacitance of 450 $pF/mm^2$. Estimate the dynamic power consumption of your chip if it has an area of 70 $mm^2$ and runs at 450 $MHz$ at $V_{DD} = 0.9$ V. |
| 3 | 20-21 2(d) | 4 | theory | Write short note on –i) Static power dissipation ii) Dynamic power dissipation. |
| 4 | 20-21 3(c) | 8 | derivation | Show that the dynamic power dissipation of a CMOS inverter varies directly with square of the power supply voltage whereas the delay of the circuit varies inversely with the power supply voltage. |
| **Leakage** | | | | |
| 5 | 19-20 5(d) | 6 | theory | Define stack effect. Find subthreshold leakage current for two series OFF transistor. |


Free shared copy · Nur Mohammad

## 4.6. Interconnect:

10 marks, 2 questions, in 1 of 2 papers. Book: Weste Ch. 6.

Most often theory (2 of 2). Subtopics: Crosstalk; Sheet resistance.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Crosstalk | **Crosstalk, floating victim** | $\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \Delta V_a$ | 19-20 4(d) |
| | Crosstalk, driven victim | $\Delta V_v = \frac{C_{adj}}{C_{gnd} + C_{adj}} \cdot \frac{1}{1+k} \Delta V_a,\ k = \frac{\tau_a}{\tau_v}$ | 19-20 4(d) |
| | Miller coupling | $C_{eff} = C_{gnd} + k_M C_{adj}$ | 19-20 4(d) |
| Sheet resistance | **Sheet resistance** | $R_\square = \frac{\rho}{t},\ R = R_\square \frac{l}{w}$ | 19-20 4(e) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Crosstalk** | | | | |
| 1 | 19-20 4(d) | 7 | theory | Define crosstalk. Show how crosstalk induces noise in different cases. How can crosstalk be reduced? |
| **Sheet resistance** | | | | |
| 2 | 19-20 4(e) | 3 | theory | Define sheet resistance. How does interconnect affect the performance of a system? |

## 4.7. Scaling:

11 marks, 2 questions, in 1 of 2 papers. Book: Weste Ch. 7.

Most often theory (2 of 2). Subtopics: Moore's law; Scaling rules.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Scaling rules | **Constant-field scaling** | $L, W, t_{ox}, V_{DD}, V_t \times \frac{1}{S};\ N_A \times S$ | 19-20 4(a) |
| | Constant-voltage scaling | $L, W, t_{ox} \times \frac{1}{S};\ V$ fixed; $N_A \times S^2$ | 19-20 4(a) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Moore's law** | | | | |
| 1 | 19-20 2(c) | 6 | theory | State Moore's law. Discuss the problems that arise due to MOSFET scaling. |
| **Scaling rules** | | | | |
| 2 | 19-20 4(a) | 5 | theory | What is scaling? Explain how different parameters of transistors are scaled? |


Free shared copy · Nur Mohammad

## 4.8. Combinational logic styles:

49 marks, 10 questions, in 2 of 2 papers. Book: Weste Ch. 9.

Most often diagram (5 of 10). Subtopics: CVSL; Static gates; Dynamic logic; Stick diagrams; Pseudo-nMOS; Pass transistors.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| CVSL | CVSL outputs | $Y = f,\ \overline{Y} = \overline{f}$ | 19-20 2(d) |
| Static gates | NAND–NAND form | $AB + CD = \overline{\overline{AB} \cdot \overline{CD}}$ | 19-20 5(c) |
| | Compound gate | $\text{AOI22} = \overline{AB + CD}$ | 19-20 5(c) |
| | Car alarm | $Y = KB\overline{D} + K\overline{B}D = K(B \oplus D)$ | 20-21 3(a) |
| Dynamic logic | Dynamic gate output | $Y = \overline{f}$ in evaluate, $Y = 1$ in precharge | 19-20 6(a) |
| Stick diagrams | **OAI22** | $\overline{f} = \overline{(A + B)(C + D)}$ | 19-20 6(b) |
| Pseudo-nMOS | Device gain factor | $k = k'\frac{W}{L},\ k' = \mu C_{ox}$ | 20-21 1(b) |
| | **Drain current, linear** | $I_D = \frac{k}{2}\left[2(V_{GS} - V_T)V_{DS} - V_{DS}^2\right]$ | 20-21 1(b) |
| | **Drain current, saturation** | $I_D = \frac{k}{2}(V_{GS} - V_T)^2$ | 20-21 1(b) |
| | Body effect | $V_T = V_{T0} + \gamma\left(\sqrt{\lvert 2\phi_F\rvert + V_{SB}} - \sqrt{\lvert 2\phi_F\rvert}\right)$ | 20-21 1(b) |
| | **Pseudo-NMOS output low** | $k_n\left[2(V_{DD} - V_{Tn})V_{OL} - V_{OL}^2\right] = k_p(V_{DD} - \lvert V_{Tp}\rvert)^2$ | 20-21 1(b) |
| | Inverter ratio | $\frac{(W/L)_n}{(W/L)_p} = \frac{k_n}{k_p} \cdot \frac{\mu_p C_{ox}}{\mu_n C_{ox}}$ | 20-21 1(b) |
| Pass transistors | 2-to-1 multiplexer | $Y = \overline{S}D_0 + SD_1$ | 20-21 6(a) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **CVSL** | | | | |
| 1 | 19-20 2(d) | 4 | theory | Explain the principle of cascode voltage switch logic. |
| **Static gates** | | | | |
| 2 | 19-20 5(c) | 9 | diagram | Design a CMOS logic circuit to compute $F = AB + CD$ using (i) only NAND gates, (ii) compound gate. Choose the transistor width to achieve unit path resistance. If each input can present a maximum of 20 $\lambda$ of transistor width, output must drive a load equivalent to 100 $\lambda$ of transistor width, calculate the minimum delay to compute F. Which one is faster? Comment on your answer. |
| 3 | 20-21 3(a) | 0 | design | You have to design a car alarm system in which the alarm will ring (Y= 1) if anyone of the following condition is satisfied. (i) The key is in (K= 1), The Seat belt is fastened (B=1) and the door is open (D=0) or (ii) The key is in, Door is closed and Seat belt is not fastened. |
| **Dynamic logic** | | | | |
| 4 | 19-20 6(a) | 6 | theory | Explain the principle of dynamic logic circuit. |


Free shared copy · Nur Mohammad

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| 5 | 20-21 2(e) | 3 | theory | What is domino gate? Draw the circuit diagram of a domino OR gate. |
| **Stick diagrams** | | | | |
| 6 | 19-20 6(b) | 6 | diagram | Draw the stick diagram of the following Boolean function: $f = (A + B).(C + D)$ with their symbolic diagram and truth table. |
| 7 | 20-21 1(c) | 6 | diagram | Draw the schematic and stick diagram of the circuit implementing the logic $Y = \overline{(A + B + C) \cdot D}$ |
| **Pseudo-nMOS** | | | | |
| 8 | 20-21 1(b) | 8 | numerical | A pseudo NMOS inverter is designed such that the output voltage becomes 0.25 V when the input voltage is 5 V. The following data are given: $\mu_n C_{ox}$ = 120 $\mu$A/V$^2$, $\mu_p C_{ox}$ = 50 $\mu$A/V$^2$, $V_{ton}$ = 1V, $V_{top}$ = -1 V, $V_{DD}$ = 5 V, $\gamma$ = 0.5. Assume that the body of the NMOS transistor is connected with the ground and the body of the PMOS transistor is connected with $V_{DD}$. (i) Draw the circuit diagram of the inverter. (ii) Calculate the inverter ratio i.e., the ratio of driver transistor (W/L) to the load transistor (W/L). (iii) What will be the output voltage if the input voltage is held at 0.25 V? Explain your answer. |
| **Pass transistors** | | | | |
| 9 | 20-21 4(d) | 3 | diagram | Implement a 2-input AND gate using both transmission gates and pass-transistor logic. |
| 10 | 20-21 6(a) | 4 | diagram | Design a 2-to-1 multiplexer using pass transistor logic. Mention its working procedure. |

## 4.9. Sequential circuits and timing:

36 marks, 6 questions, in 2 of 2 papers. Book: Weste Ch. 10.

Most often theory (2 of 6). Subtopics: Hold time; Cycle time; Setup and hold; Flip-flops; FSM.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Hold time | **Min-delay (hold)** | $t_{cd} \geq t_{hold} - t_{ccq}$ | 19-20 5(a) |
| | Hold with skew | $t_{cd} \geq t_{hold} - t_{ccq} + t_{skew}$ | 19-20 5(a) |
| Cycle time | **Max-delay (setup)** | $T_c \geq t_{pcq} + t_{pd} + t_{setup}$ | 19-20 5(b) |
| | Itanium 2 bypass | $T_c = 90 + 1010 + 62 = 1162$ ps | 19-20 5(b) |
| Setup and hold | Latest data arrival | $t_{data,max} = t_{clk1,max} + t_{clk\text{-}q,max} + t_{logic,max}$ | 20-21 4(a) |
| | **Setup constraint** | $t_{data,max} + t_{setup} \leq T + t_{clk2,min}$ | 20-21 4(a) |
| | **Hold constraint** | $t_{data,min} \geq t_{clk2,max} + t_{hold}$ | 20-21 4(a) |
| | Maximum frequency | $f_{max} = \dfrac{1}{T_{min}}$ | 20-21 4(a) |
| FSM | Divide-by-3 next state | $S_1' = S_0,\ S_0' = \overline{S_1 + S_0},\ y = \overline{S_1}\,\overline{S_0}$ | 20-21 6(c) |

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Hold time** | | | | |
| 1 | 19-20 5(a) | 6 | theory | What is min-delay constraint? Find min-delay for a system sequenced with flip-flop. If min-delay occurs, what is the solution? |
| **Cycle time** | | | | |
| 2 | 19-20 5(b) | 4 | numerical | The Itanium 2 ALU self-bypass path is shown in Figure 5.1, and delays associated with each element are given in Table 5.1. Calculate the minimum cycle time $T_c$ at which ALU self-bypass path will operate correctly. The earliest input to the late bypass multiplexer is imm value coming from another flip-flop. Will this path experience any hold-time failures? |
| **Setup and hold** | | | | |
| 3 | 20-21 4(a) | 10 | numerical | In the following circuit of Figure 4(a), determine whether there is any set up or hold time violation. |
| **Flip-flops** | | | | |
| 4 | 20-21 4(b) | 6 | diagram | Show the circuit diagram of an asynchronous resettable Flip-Flop and explain its operation utilizing a timing diagram. |
| 5 | 20-21 6(f) | 4 | theory | Fill in the truth table for the flip flop shown in Figure 6(f). Is this a positive edge triggered flip flop? |
| **FSM** | | | | |
| 6 | 20-21 6(c) | 6 | design | You have to design a divide-by-3 counter using FSM. (i) Draw the state transition diagram of the finite state machine described. (ii) Derive the state transition table and equations for the next-state logic. (iii) Draw the complete FSM circuit. |

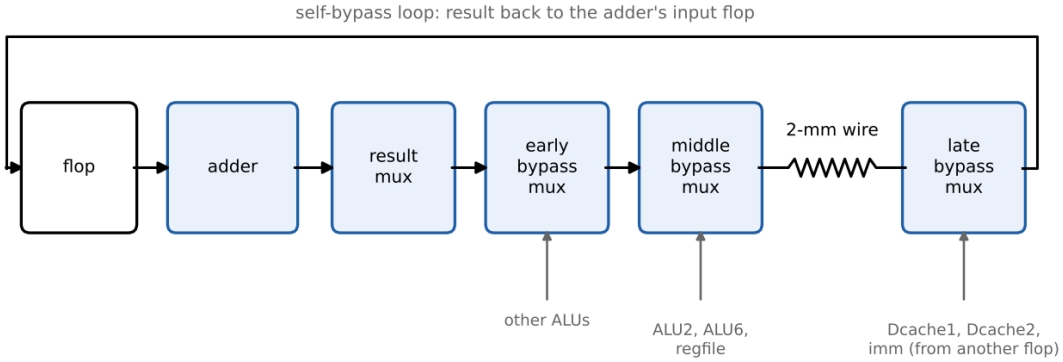



**Figure 6:** 19-20 5(b): Figure 5.1 of the paper (redrawn)

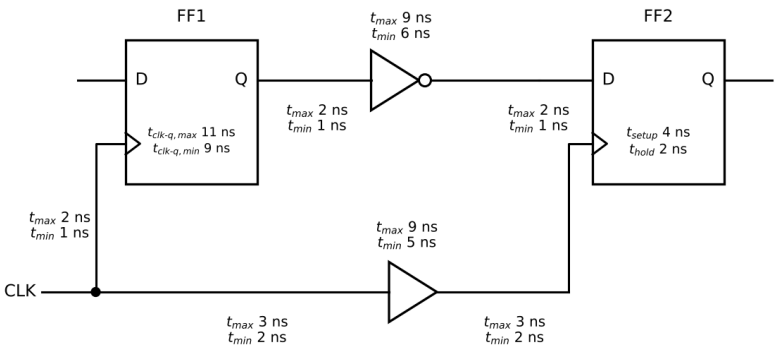



**Figure 7:** 20-21 4(a): Figure 4(a) of the paper (redrawn)


Free shared copy · Nur Mohammad

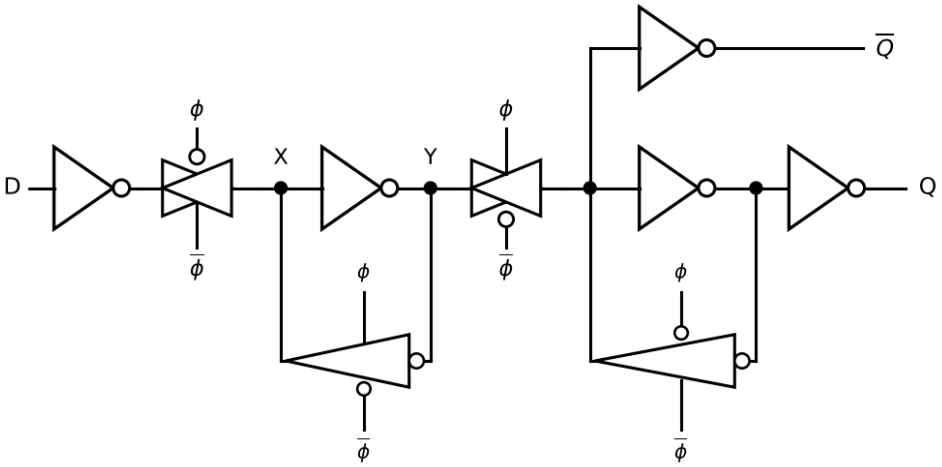



**Figure 8:** 20-21 6(f): Figure 6(f) of the paper (redrawn)

| $\phi$ | $\overline{\phi}$ | D | X | Y | $\overline{Q}$ | Q |
|---|---|---|---|---|---|---|
| 0 | 1 | | 1 | | | 1 |
| 1 | 0 | 1 | | | | |
| 0 | 1 | 1 | | | | |
| 1 | 0 | 1 | | | | |

**Figure 9:** 20-21 6(f): The table of Figure 6(f), as printed

## 4.10. Datapath:

7 marks, 1 question, in 1 of 2 papers. Book: Weste Ch. 11.

Most often theory (1 of 1). Subtopics: Bus architectures.

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Bus architectures** | | | | |
| 1 | 20-21 5(b) | 7 | theory | A 4-bit datapath consists of register, ALU and a shifter. Show two possible bus architectures of the system such that an addition operation of two operands stored in the register and storing the result back in the register can be computed in at most two clock cycles. |


Free shared copy · Nur Mohammad

## 4.11. Memories and arrays:

26 marks, 4 questions, in 2 of 2 papers. Book: Weste Ch. 12.

Most often diagram (3 of 4). Subtopics: SRAM; ROM; PLA.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| SRAM | SRAM sizing | pull-down > access > pull-up | 19-20 6(c) |
| | Bit-line read time | $t = \frac{C_{bit}\,\Delta V}{I_{cell}}$ | 20-21 5(a) |
| ROM | **NOR ROM** | bit line $= \overline{\sum \text{word lines with a transistor}}$ | 19-20 6(d) |
| PLA | XOR as sum of products | $A \oplus B = A\overline{B} + \overline{A}B$ | 20-21 6(e) |

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **SRAM** | | | | |
| 1 | 19-20 6(c) | 6 | theory | Draw logic diagram of a SRAM cell and explain its different modes of operation. |
| 2 | 20-21 5(a) | 10 | diagram | (i) Show the circuit diagram of a 2 x 2 SRAM array which uses 6-transistor memory cell as basic storage cell. Clearly show the row select, column select, pre-charge and sense signal in your circuit. (ii) Explain how the READ and WRITE operation are performed showing the second row and first column cell as an example. (iii) Explain how sense amplifier reduces the READ time of the cell. |
| **ROM** | | | | |
| 3 | 19-20 6(d) | 7 | diagram | Design and implement a pseudo-nMOS ROM to store the data shown in Figure 6.1. Also draw the dot diagram representation. |
| **PLA** | | | | |
| 4 | 20-21 6(e) | 3 | diagram | Sketch a dot diagram for a 2-input XOR using a PLA. |

## 4.12. Verilog and design flow:

14 marks, 3 questions, in 1 of 2 papers. Book: Weste Appendix A.

Most often design (2 of 3). Subtopics: Design flow; Verilog.

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Design flow** | | | | |
| 1 | 20-21 4(c) | 6 | theory | Suppose you have written Verilog code to operate a microprocessor. What will be the upcoming steps till its fabrication? Briefly explain your answer. |
| **Verilog** | | | | |
| 2 | 20-21 6(b) | 5 | design | Design a 7-segment display driver in Verilog. |
| 3 | 20-21 6(d) | 3 | design | Write a Verilog code for the FSM described in Question 6(c). |


Free shared copy · Nur Mohammad

# 5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

| No. | Paper | Marks | Derivation | Outline |
|---|---|---|---|---|
| **MOS transistor theory** | | | | |
| 1 | 19-20 2(a) | 8 | I-V equations | Write the channel charge and electron speed; integrate along the channel; set the pinch-off condition. |
| **Power** | | | | |
| 2 | 20-21 3(c) | 8 | Dynamic power | Add the energy per cycle for dynamic power; divide the capacitor charge by the saturation current for delay. |

# 6. Repeat tracker:

No repeats among these papers.

# 7. Question style:

How each topic was asked, as a number of questions.

| Topic | Theory | Derivation | Numerical | Diagram | Design |
|---|---|---|---|---|---|
| MOS transistor theory | 2 | 1 | · | · | · |
| CMOS inverter | 3 | · | · | · | · |
| Fabrication | 6 | · | · | · | · |
| Delay and logical effort | 1 | · | 5 | · | 2 |
| Power | 3 | 1 | 1 | · | · |
| Interconnect | 2 | · | · | · | · |
| Scaling | 2 | · | · | · | · |
| Combinational logic styles | 3 | · | 1 | 5 | 1 |
| Sequential circuits and timing | 2 | · | 2 | 1 | 1 |
| Datapath | 1 | · | · | · | · |
| Memories and arrays | 1 | · | · | 3 | · |
| Verilog and design flow | 1 | · | · | · | 2 |

# 8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

| No. | Topic | Marks | Covered so far |
|---|---|---|---|
| 1 | Combinational logic styles | 49 | 17 % |
| 2 | Delay and logical effort | 46 | 32 % |
| 3 | Sequential circuits and timing | 36 | 45 % |
| 4 | Fabrication | 30 | 55 % |
| 5 | Power | 28 | 64 % |
| 6 | Memories and arrays | 26 | 73 % |
| 7 | CMOS inverter | 20 | 80 % |
| 8 | MOS transistor theory | 17 | 86 % |
| 9 | Verilog and design flow | 14 | 90 % |
| 10 | Scaling | 11 | 94 % |
| 11 | Interconnect | 10 | 98 % |
| 12 | Datapath | 7 | 100 % |


Free shared copy · Nur Mohammad